3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | WARM UPS | RETRIEVAL BEFORE NEW EXAMPLES

Name: _____ Date: _____

1 | Multiple Choice NO CALCULATOR

Recent review | 3.8 Inverse Functions

Let $f(x) = x^3 + 2x + 3$. Which equation is the tangent line to the inverse function f^{-1} at $(6, 1)$?

A. $y - 1 = 5(x - 6)$

B. $y - 1 = \frac{1}{5}(x - 6)$

C. $y - 6 = \frac{1}{5}(x - 1)$

D. $y - 1 = -\frac{1}{5}(x - 6)$

E. $y - 6 = 5(x - 1)$

Show your reasoning here.

2 | Multiple Choice CALCULATOR PERMITTED

Spiral review | Motion and Tangent Lines

A particle has position $s(t) = 0.6t^3 + 2\sin(0.9t)$ meters. Which equation approximates the tangent line at $t = 1.6$ seconds? Use radians.

A. $s - 4.84 = 4.44(t - 1.6)$

B. $s - 4.44 = 2.88(t - 1.6)$

C. $s - 4.44 = -4.84(t - 1.6)$

D. $s - 4.44 = 4.84(t - 1.6)$

E. $s - 3.61 = 4.84(t - 1.6)$

Show your reasoning here.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | LEARNING FOCUS & VOCABULARY | READ, HIGHLIGHT, CONNECT

Learning focus | I can...

- Differentiate exponential functions $e^{u(x)}$ and $a^{u(x)}$ using the Chain Rule.
- Differentiate natural-log and other logarithmic functions using $(\ln u)' = u'/u$.
- Use implicit differentiation to explain why logarithm and exponential rules work.
- Apply logarithm properties and logarithmic differentiation to products, quotients, and powers.

Vocabulary | Highlight the important words

Exponential function

A function where the variable appears in the exponent, such as e^x or 3^x.

Natural logarithm

The inverse of e^x, written $\ln x$; its domain is $x > 0$.

Base

The positive constant a in a^x; for logarithms, $a \neq 1$.

Logarithm

An exponent in another form: $\log_b a = c \iff b^c = a$.

Chain Rule

Multiply the outside derivative by the derivative of the inside function.

Change of base

Rewrite $\log_b x = \frac{\ln x}{\ln b}$ to differentiate.

Logarithm properties

Product, quotient, and power rules that rewrite products as sums and powers as factors.

Logarithmic differentiation

Take natural logs, simplify, differentiate implicitly, and solve for the derivative.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | MEANING & METHOD | REFERENCE NOTES: REPEAT BOTH LESSONS

1 | Exponential functions and the Chain Rule

$\frac{d}{dx}e^x = e^x \quad \frac{d}{dx}e^{u(x)} = e^{u(x)}u'(x) \quad \frac{d}{dx}a^{u(x)} = a^{u(x)}\ln(a)\,u'(x)$

2 | Natural logarithm and other bases

$\frac{d}{dx}\ln x = \frac{1}{x} \quad \frac{d}{dx}\ln(u) = \frac{u'}{u} \quad \frac{d}{dx}\log_b x = \frac{1}{x\ln b}$

Use the Chain Rule for an inside expression. For $\ln(u)$, the logarithm requires $u > 0$.

3 | Properties of logarithms

$\log_b(MN) = \log_b M + \log_b N \quad \log_b(M/N) = \log_b M - \log_b N \quad \log_b(M^k) = k\log_b M \quad \log_b a = \frac{\ln a}{\ln b}$

For these identities, the logarithm arguments must be positive. The base must satisfy $b > 0,\ b \neq 1$.

4 | Logarithmic differentiation method

1. Begin with $y = \text{product} \,/\, \text{quotient} \,/\, \text{power}$. Take $\ln$ of both sides.
2. Use log properties to bring down exponents and split products/quotients.
3. Differentiate implicitly: the left side becomes $\frac{y'}{y}$.
4. Multiply by y and substitute the original expression back.

Notes / questions:

3.9 | Exponential and Logarithmic Functions

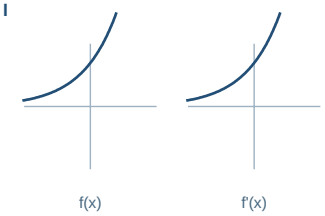
AP CALCULUS AB | LESSON 1 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 1 AND A MATCHED PRACTICE

Example 1 | Graph relationships

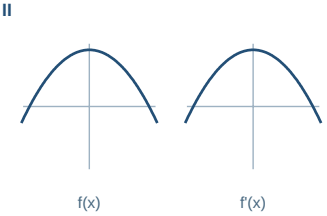
From Section 3.9 source notes

Which of the following pairs of graphs could represent a function and its derivative?

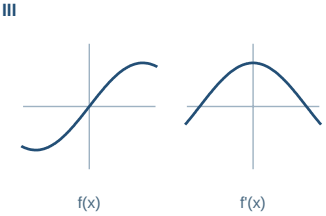
I



II



III



A. I only B. II only C. III only D. I and III only E. II and III only

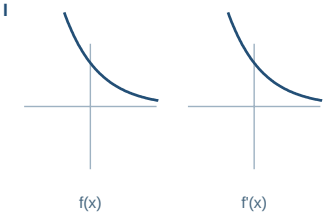
Show work and justify each step.

You Try It 1 | Graph relationships

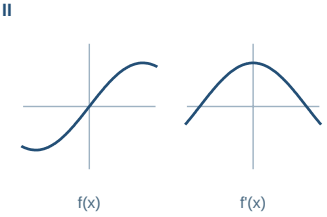
Matching practice (new numbers)

For the three new graph pairs, decide which could represent a function and its derivative.

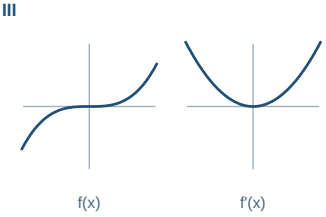
I



II



III



A. I only B. II only C. III only D. I and III only E. II and III only

Show work and justify each step.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 2 AND A MATCHED PRACTICE

Example 2 | The exponential Chain Rule

From Section 3.9 source notes

Find $\frac{d}{dx}\left(e^{2x-1}\right)$.

Show work and justify each step.

You Try It 2 | The exponential Chain Rule

Matching practice (new numbers)

Find $\frac{d}{dx}\left(e^{3x-2}\right)$.

Show work and justify each step.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 3 AND A MATCHED PRACTICE

Example 3 | Exponential with a rational exponent

From Section 3.9 source notes

Find $\frac{d}{dx}\left(e^{-3/x}\right)$.

Show work and justify each step.

You Try It 3 | Exponential with a rational exponent

Matching practice (new numbers)

Find $\frac{d}{dx}\left(e^{-5/x}\right)$.

Show work and justify each step.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 4 AND A MATCHED PRACTICE

Example 4 | Derive the natural-log rule

From Section 3.9 source notes

Use implicit differentiation to prove $\frac{d}{dx}\ln x = \frac{1}{x}$ for $x > 0$.

Show work and justify each step.

You Try It 4 | Derive the natural-log rule

Matching practice (new numbers)

Use implicit differentiation to show the derivative of $y = \ln(3x)$ for $x > 0$.

Show work and justify each step.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 5 AND A MATCHED PRACTICE

Example 5 | Natural logarithm with the Chain Rule

From Section 3.9 source notes

Let $y = \ln(2x + 2)$. Find y'.

Show work and justify each step.

You Try It 5 | Natural logarithm with the Chain Rule

Matching practice (new numbers)

Let $y = \ln(3x + 5)$. Find y'.

Show work and justify each step.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 1 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 6 AND A MATCHED PRACTICE

Example 6 | Natural log of a trigonometric function

From Section 3.9 source notes

Let $f(x) = \ln(\tan x)$. Find $f'(x)$.

Show work and justify each step.

You Try It 6 | Natural log of a trigonometric function

Matching practice (new numbers)

Let $f(x) = \ln(\tan(2x))$. Find $f'(x)$.

Show work and justify each step.